

Clinical Engineering Program: Equipment Maintenance and AEM Procedure

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1. Purpose and scope

This procedure describes how Clinical Engineering inventories, classifies, maintains and repairs medical equipment at Tallowbrook facilities, and how maintenance performance is measured and reported.

It applies to medical equipment at both hospitals, Outpatient Pavilion West, Imaging & Therapy Stafford and Urgent & Family Care Mission Bend. Surgery Center Grand Parkway devices are covered under its services agreement.

2. Organization

Clinical Engineering has 13 FTEs: a director, two senior specialists, nine technicians and a coordinator. Shops are located at Medical Center West and at the System Service Center in Stafford, which supports the Fort Bend sites.

Imaging equipment and medical lasers are serviced under manufacturer contracts. Clinical Engineering tracks these devices in inventory, schedules vendor visits and reviews vendor service reports.

3. Equipment inventory

Every medical device must pass an incoming inspection and be entered in the CMMS before clinical use. Inventory records are kept in two systems until consolidation:

- Legacy CE CMMS: Medical Center West and the Pavilion (in use since 2012).
- Enterprise CMMS: Hospital Fort Bend, the surgery center, Stafford and Mission Bend (since 2020). Facilities work orders are also in this system.

4. Risk classification and critical equipment

Each device type is assigned high, medium or low risk based on its function, the consequence of failure and its maintenance requirements. The assignment is made by a senior specialist and recorded as a category default in the CMMS.

Critical equipment is equipment whose failure could result in serious injury or death to patients or staff. All critical equipment is classified high risk and is maintained according to the manufacturer's recommendations.

5. Maintenance strategies

Federal requirements allow a hospital to adjust maintenance activities and intervals from manufacturer recommendations, based on a documented risk assessment by qualified personnel.

Tallowbrook's alternate equipment maintenance (AEM) program applies this to eligible lower-risk devices.

- Manufacturer schedule required: imaging and radiologic equipment, medical lasers, new devices without sufficient maintenance history, and high-risk life-support devices.
- AEM eligible: other medium- and low-risk devices where the risk assessment supports a different interval or scope.
- AEM decisions are recorded by device category and reviewed at least annually, or sooner after an incident or recall.

6. Work order handling

Scheduled maintenance (PM), corrective repairs, incoming inspections and recall or alert actions are all recorded as work orders. The steps differ slightly between the two systems.

- Legacy CE CMMS: PM work orders are generated on the first of each month for the month. Requests for repair are phoned or emailed to the shop and entered by the coordinator.
- Enterprise CMMS: PM work orders are generated 14 days before the due date. Staff submit repair requests through the portal, which also routes facilities requests.
- In both systems a technician records labor hours, parts and the result, and closes the work order. A repair call where no fault is found is closed as the no problem found type.

7. Recalls and alerts

The coordinator reviews manufacturer and regulatory notices weekly, searches both systems for affected models and opens a recall or alert work order for each affected device. Unlocated devices are reported to the director within five business days.

8. Performance measurement

Scheduled maintenance performance is the share of PM work orders completed by their due date. It is reported monthly to the Environment of Care committee overall, for high-risk devices and by CMMS instance.

The target is 95% overall and 100% for high-risk devices. Any overdue high-risk PM is escalated to the director the day it becomes overdue.